

Brent Crude Under OPEC+ Production Discipline

Stochastic Differential Equation — Derivation and Simulation Framework

Geeconomic SDE Project · Case 1

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1 Observation and Motivation

Brent crude does not random-walk indefinitely. Over medium horizons (weeks to months) it exhibits persistent mean-reversion toward a structural band of approximately \$70–\$90. This reversion is not statistical — it is enforced by real production decisions made by two actors with identifiable and stable incentive structures.

The question this model answers is:

Given Brent currently at \$82, what is the full probability distribution of price at 30, 60, and 90 days from now, under the structural constraints imposed by OPEC+ and US shale?

This is not a point forecast. The output is a probability distribution over price levels at each horizon — directly tradeable against market option pricing.

2 Economic Hypothesis

2.1 The Two-Actor Band Mechanism

Floor (\$70–75): OPEC+ member fiscal breakevens cluster near this range (Saudi Arabia $\approx$ \$70, Iraq $\approx$ \$75). Below it, members run budget deficits and have a strong collective incentive to cut production. The cut is a *rate* decision — supply is reduced continuously until price recovers. Crucially, the response is proportional: a small breach triggers modest cuts; a large breach triggers coordinated emergency response (as documented in 2020 and 2022 cut cycles).

Ceiling (\$85–90): Above this range, US shale operators can profitably hedge forward production. Rig counts rise and new supply enters within 3–6 months, continuously capping upside. Again the response is proportional — higher prices bring more rigs online, not a binary switch.

The proportionality of both responses is what makes price *mean-reverting* rather than merely bounded. There is no wall that price bounces off — there is a restoring force that grows stronger the further price moves from the band center.

2.2 Why Volatility Is State-Dependent

Market uncertainty is not uniform across the price band. Near \$70 and \$90, OPEC meetings, shale hedging reports, and inventory data are watched closely — implied volatility is elevated, realized vol is high. In the middle of the band (\$76–\$84), markets are calm and vol is low.

Therefore σ must be a function of distance from θ , not a constant. This is the key nonlinearity and the primary source of analytical intractability.

2.3 Jump Sources

Geopolitical events arrive discretely, not continuously: surprise OPEC+ production decisions, tanker route disruptions (e.g. Houthi attacks), sudden China demand revisions from PMI data. A continuous diffusion term alone cannot replicate these — an explicit jump component is required.

2.4 State Variable Choice

Let $X(t) = \log(\text{Brent spot price at time } t)$. We work in log-price rather than price for two reasons: (i) price cannot go negative, and (ii) percentage moves are the economically meaningful unit — a \$5 move at \$40 is very different from \$5 at \$80.

3 SDE Derivation

3.1 Building the Expression Term by Term

Step 1 — Drift. The restoring force argument says: the rate of change of X is pulled toward θ at a speed proportional to how far it currently is from θ .

“If price is above target, drift is negative. If below, drift is positive. The further away, the stronger the pull.”

$$\mu(X, t) = \kappa(\theta - X) \quad (1)$$

This is the Ornstein-Uhlenbeck drift. κ controls reversion speed. A value of $\kappa = 2.1 \text{ yr}^{-1}$ implies a half-life of $\ln(2)/\kappa \approx 4$ months, consistent with observed OPEC response cycles.

Step 2 — Diffusion. Constant σ would assert that uncertainty is uniform across all price levels. Our mechanism says otherwise. We encode this as:

$$\sigma(X, t) = \sigma_0 \cdot (1 + \alpha |X - \theta|) \quad (2)$$

where σ_0 is baseline annualised volatility and α controls how quickly vol scales with distance from the mean. Minimum uncertainty is at $X = \theta$; it rises as price deviates in either direction.

Step 3 — Jump. Geopolitical shocks arrive as a Poisson process with intensity λ (average events per year). Each shock moves log-price by $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$. This is a compound Poisson process:

$$dq_t : \quad P(dq = 1) = \lambda dt, \quad J \sim \mathcal{N}(\mu_j, \sigma_j^2) \quad (3)$$

3.2 Full SDE

Assembling the three components:

$$\boxed{dX = \kappa(\theta - X) dt + \sigma_0(1 + \alpha|X - \theta|) dW + J dq} \quad (4)$$

Term	Type	Economic meaning
$\kappa(\theta - X) dt$	Drift	OPEC/shale restoring force
$\sigma_0(1 + \alpha X - \theta) dW$	Diffusion	State-dependent market uncertainty
$J dq$	Jump	Discrete geopolitical shock

4 Analytical Tractability Check

Before building a numerical solver, we attempt the analytical path and identify exactly where it fails. This is the most important intellectual step — it justifies the numerical approach rigorously.

Q1: Is the diffusion coefficient constant?

No. $\sigma(X) = \sigma_0(1 + \alpha|X - \theta|)$ is nonlinear in X . The term $|X - \theta|$ has derivative $\text{sign}(X - \theta)$ — a step function, discontinuous at $X = \theta$.

Applying Itô's lemma to any transformation $f(X)$ requires computing:

$$df = f'(X) dX + \frac{1}{2} f''(X) (dX)^2 \quad (5)$$

The quadratic variation term produces $\frac{1}{2} f''(X) \cdot \sigma(X)^2 dt$. With $\sigma(X)$ nonlinear, $\sigma^2 = \sigma_0^2(1 + \alpha|X - \theta|)^2$ is a nonlinear function of a stochastic process. The resulting Fokker-Planck PDE:

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x} [\kappa(\theta - x) p] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma_0^2(1 + \alpha|x - \theta|)^2 p] \quad (6)$$

has a non-smooth diffusion coefficient and admits no closed-form transition density.

→ **FAILS at Q1. Analytical solution does not exist.**

Q2: Is drift linear in X ? Yes — $\kappa(\theta - X)$ is linear. This alone (classical OU) would permit an analytic solution, but Q1 already failed.

Q3: Are jumps tractable in isolation? λ is constant and $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$, so Merton's formula would apply to the jump component alone. Irrelevant here — Q1 already forces numerical treatment.

Decision: Euler-Maruyama numerical solver required.

5 Numerical Solution: Euler-Maruyama

5.1 Discretisation

Over time step $\Delta t = 1/252$ (one trading day):

$$X(t + \Delta t) = X(t) + \kappa(\theta - X(t)) \cdot \Delta t + \sigma_0(1 + \alpha|X(t) - \theta|) \cdot \sqrt{\Delta t} \cdot Z + J \cdot \mathbf{1}_{[\text{Poisson}(\lambda \Delta t) \geq 1]} \quad (7)$$

where $Z \sim \mathcal{N}(0, 1)$ and $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$. This is the exact update rule implemented in `core/euler_maruyama.py`.

5.2 Implementation Architecture

The solver follows a model-injection pattern. The core loop in `euler_maruyama.py` calls three functions at each step:

Function	Implements
<code>model.drift(X, t)</code>	$\kappa(\theta - X)$
<code>model.diffusion(X, t)</code>	$\sigma_0(1 + \alpha X - \theta)$
<code>model.jump(X, t, rng, dt)</code>	Compound Poisson draw

Each case only overrides these three functions. The solver loop is shared across all five cases in the project.

6 Parameters

Parameter	Value	Units	Reasoning
κ	2.1	yr ⁻¹	≈4 month half-life, matches OPEC response cycle
θ	log(78)	—	Midpoint of OPEC+ fiscal breakeven band
σ_0	0.18	—	Baseline annualised vol, historical Brent
α	0.40	—	Vol scaling near band boundaries
λ	3.0	yr ⁻¹	≈3 material geopolitical shocks per year
μ_j	0.0	—	Shocks symmetric on average
σ_j	0.06	—	Typical ±6% log-price move per shock event

Parameters are currently set by economic reasoning. Formal MLE calibration to historical Brent data is to be implemented in `calibration/estimate.py`.

7 Results and Inference

7.1 Simulation Setup

$X_0 = \log(82)$, horizon $T = 0.5$ years, $\Delta t = 1/252$, $N = 10,000$ Monte Carlo paths.

7.2 Scenario Probability Table

Horizon	P(<\$70)	P(>\$90)	P(in band)	Median	5th pct	95th pct
30d	1.3%	5.4%	93.4%	\$81.0	\$73.2	\$90.3
60d	4.7%	8.7%	86.7%	\$80.3	\$70.3	\$92.1
90d	7.0%	9.3%	83.7%	\$79.7	\$68.8	\$93.1

7.3 Reading the Output

Median path. Moves $\$82 \rightarrow \$81.0 \rightarrow \$80.3 \rightarrow \79.7 , drifting toward $\theta = \$78$. Mean-reversion is working as specified: starting above target, the drift term $\kappa(\theta - X)$ is negative at every step.

Asymmetry. At 30 days, $P(>\$90) = 5.4\%$ versus $P(<\$70) = 1.3\%$. Starting at $\$82$, the ceiling is geometrically closer than the floor in log-price terms — less movement required to breach $\$90$ than $\$70$. This asymmetry is structural, not assumed.

Tail widening. The band probability decays from 93.4% to 83.7% over 90 days as jump events and state-dependent diffusion accumulate tail paths over time.

7.4 Tradeable Inference

Standard Black-Scholes options pricing uses a single implied volatility number and a log-normal assumption. It has no concept of structural levels at \$70 and \$90 and no mean-reversion term. It therefore prices equidistant strikes approximately symmetrically.

Our model says the ceiling breach (5.4%) is $\approx 4\times$ more likely than the floor breach (1.3%) at 30 days, purely from starting position and structural geometry. If the options market is pricing \$90 calls and \$70 puts at similar implied volatilities — treating both tails as equally probable — then \$90 calls are underpriced relative to our model.

The trade: long \$90 calls, short \$70 puts. The thesis has a timing constraint — by 90 days the two tail probabilities converge (7.0% vs 9.3%) and the skew argument weakens. The model specifies both the direction and the horizon of the edge.

8 Simulation Output

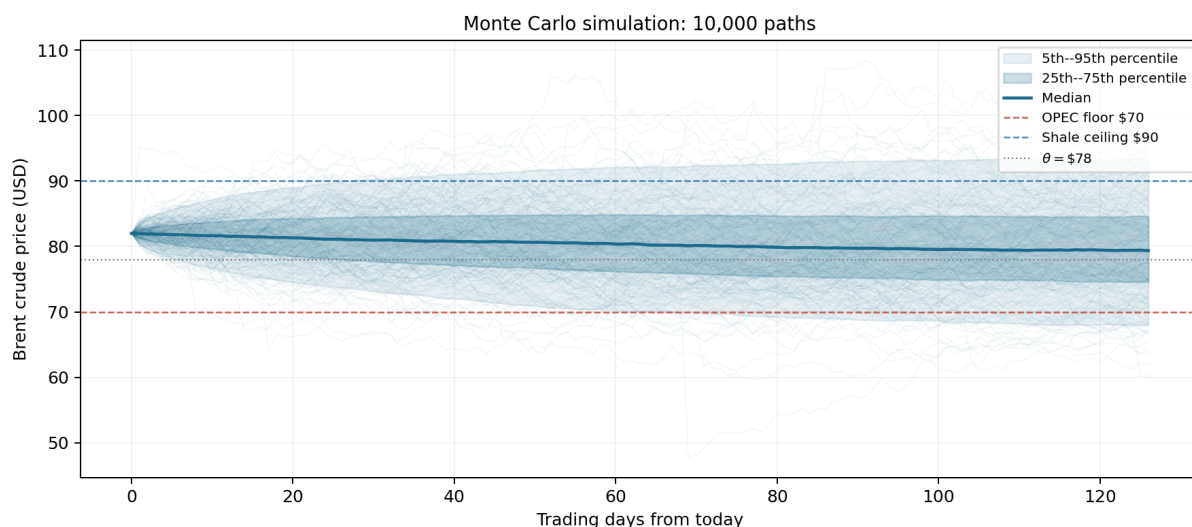


Figure 1: 10,000 Monte Carlo paths for Brent crude, converted to price space. Shaded bands show the 5th–95th and 25th–75th percentile ranges. The median path drifts from \$82 toward $\theta = \$78$ as mean-reversion takes effect. Dashed lines mark the OPEC floor (\$70) and shale ceiling (\$90). Occasional sharp single-step moves in individual paths are jump events.

9 Project Context

This is Case 1 of a five-case Geoeconomic SDE framework. Each case introduces one new numerical technique on top of the last:

Case	Asset / Scenario	New technique introduced
1	Brent crude under OPEC+	Euler-Maruyama, state-dependent diffusion
2	EM FX under sanctions shock	Poisson jump process, compound arrivals
3	Yield curve $\leftrightarrow$ equity	Cholesky decomposition, correlated SDEs
4	Commodity regime switching	Markov chain, conditional parameter switching
5	Geoeconomic scenario engine	Full 4D Monte Carlo, scenario scoring

Repository: `geoeconomic-sde/cases/01_oil_pec/` · *Core solver:* `core/euler_maruyama.py` · *Parameters:* `calibration/params.json`